

TumorAI XAI — Comprehensive System Architecture, Features & Technical Specification

- > **Project Name:** TumorAI XAI — Persistent Patient History & Secure Medical Report Platform
- > **Domain:** Deep Learning, Explainable AI (XAI), Radiology Diagnostics, Healthcare IT Interoperability
- > **Version:** 2.5.0 (Advanced Intelligence Suite Edition)
- > **Tech Stack:** Python (Flask), TensorFlow / Keras, Supabase (PostgreSQL & Storage), OpenCV, PyDICOM, ReportLab, Google Gemini AI, HL7 FHIR R4 Standard, Vanilla HTML5/CSS3/JavaScript.

Executive Summary

TumorAI XAI is an enterprise-grade Explainable Artificial Intelligence (XAI) clinical decision-support system designed for radiology departments and medical research. The platform provides end-to-end automation for brain tumor detection, classification, explainability visualization, epistemic uncertainty measurement, DICOM processing, patient study management, FHIR R4 interoperability, automated clinical report generation, and an **Advanced Intelligence Suite** comprising 3D tumor volume segmentation, longitudinal growth tracking, counterfactual sensitivity analysis, AI reliability indexing, pre-inference data quality auditing, study-grounded Clinical AI Copilot Q&A, multi-planar 3D rendering, and patient digital journey tracking.

The application strictly adheres to enterprise medical security standards by storing raw DICOM/MRI images, saliency heatmaps, binary region-of-interest (ROI) masks, and PDF reports in **private storage buckets** accessible exclusively via short-lived signed URLs (300-second TTL), keeping credentials strictly on the backend.

1. Advanced Intelligence Suite & Signature Features

TumorAI XAI includes **8 signature clinical features** built on top of the core ensemble and explainability engines:

1.1 ■ Automatic Tumor Segmentation & 3D Volume Calculation (`modules/segmentation_engine.py`)

- **Mechanism:** OpenCV spatial contour extraction and morphological filtering on high-intensity lesion regions.
- **3D Volumetric Estimation:** Computes estimated 3D tumor volume ($V = \frac{4}{3} \pi \cdot a \cdot b \cdot c$) in cm^3 , maximum linear diameter (cm), cross-sectional area (cm^2), bounding box, and anatomical quadrant location.
- **Boundary Contour Overlay:** Renders transparent colored boundary maps outlining lesion boundaries for quick visual review.

1.2 ■ Longitudinal Tumor Growth Tracker & Treatment Response Monitor (`services/longitudinal_service.py`)

- **Volumetric Trajectory:** Tracks volume changes ($\Delta V = V_{\text{latest}} - V_{\text{baseline}}$) and percentage growth rates across historical MRI studies for a patient.
- **Monthly Growth Velocity:** Computes average percentage volume change per month ($\% / \text{month}$).
- **RANO / RECIST Treatment Criteria:** Categorizes patient response into:
- **Complete Response (CR):** Tumor undetectable (100% regression).

- **Partial Response (PR):** Volume shrinkage $\leq -50\%$.
- **Stable Disease (SD):** Controlled volume between -50% and $+25\%$.
- **Progressive Disease (PD):** Tumor growth $> +25\%$.

1.3 ■ Multi-Planar & 3D Tumor Mesh Viewer

- **Orthogonal Plane Rendering:** View synchronized Orthogonal MRI Slices across **Axial**, **Coronal**, and **Sagittal** planes.
- **Interactive 3D Tumor Mesh:** Canvas 2D/3D surface mesh visualization allowing radiologists to rotate and inspect 3D lesion geometry.

1.4 ■ AI Reliability Panel & Model Agreement Matrix

- **Overall AI Reliability Index (0 - 100%):** Synthesizes model confidence (35%), XAI spatial alignment (25%), data quality (20%), and model consensus (20%).
- **Model Agreement Visual Matrix:** Displays individual predictions, confidence levels, and agreement status across **VGG16**, **ResNet50**, and **EfficientNetB0**.

1.5 ■ Counterfactual Explainable AI & Sensitivity Analysis (`modules/counterfactual_engine.py`)

- **Contrastive Region Modification:** Evaluates minimal spatial region perturbations required to alter prediction boundaries.
- **Sensitivity Metrics:** Computes prediction sensitivity scores and highlights top influential spatial regions on contrastive heatmaps.

1.6 ■ Pre-Inference AI Data Quality & OOD Checker (`modules/quality_checker.py`)

- **Pre-Flight Image Audit:** Automatically audits incoming MRI scans prior to deep learning model execution:
- Resolution Check ($\geq 128 \times 128$)
- Mean Brightness (15.0 - 235.0)
- Contrast Check (Standard deviation of pixel intensities)
- Sharpness / Blur Audit (Laplacian variance ≥ 45.0)
- Out-Of-Distribution (OOD) score estimation based on border profile and pixel entropy.

1.7 ■ Study-Grounded Clinical AI Copilot (`services/copilot_service.py`)

- **Interactive Radiologist Assistant:** Powered by Google Gemini AI, answering radiologist queries strictly grounded in the active patient record, scan findings, XAI metrics, 3D volume data, and longitudinal history.

1.8 ■ Digital Patient Journey Timeline

- **Timeline Navigation:** Visual milestone timeline linking initial MRI, follow-up scans, AI tumor detection, volume delta history, and generated PDF reports.

2. Machine Learning Models & Inference Architecture

The system supports individual model inference as well as a multi-architecture ensemble voting engine for robust classification across four distinct diagnostic categories:

- [glioma](#) — Glioma tumor detected
- [meningioma](#) — Meningioma tumor detected
- [pituitary](#) — Pituitary tumor detected
- [notumor](#) — No brain tumor detected (Healthy MRI)

2.1 Neural Network Backbones

1. VGG16 Transfer Learning (Primary Model)

- **Architecture:** 16-layer Deep Convolutional Neural Network pre-trained on ImageNet with custom dense classification head.
- **Input Dimensions:** `128 x 128 x 3` (normalized RGB imaging tensor).
- **Target Feature Layer for XAI:** `block5_conv3` (final convolutional block capturing high-level spatial semantic features).
- **Weight File:** `models/vgg16_best.h5` (or fallback `models/model.h5`).

2. ResNet50 Backbone

- **Architecture:** 50-layer Residual Network with skip connections to address vanishing gradients and capture multi-scale feature representations.
- **Weight File:** `models/resnet50_best.h5`.

3. EfficientNetB0 Backbone

- **Architecture:** Scaled convolutional neural network optimizing computational efficiency and parameter utilization.
- **Weight File:** `models/efficientnetb0_best.h5`.

2.2 Multi-Model Ensemble & Consensus Engine (`modules/ensemble_engine.py`)

To eliminate single-model bias and ensure high clinical confidence, TumorAI XAI incorporates a weighted soft-voting ensemble mechanism:

- **Weighted Probability Aggregation:**

$$P_{\text{ensemble}} = 0.50 \cdot P_{\text{VGG16}} + 0.25 \cdot P_{\text{ResNet50}} + 0.25 \cdot P_{\text{EfficientNetB0}}$$

- **Inter-Model Consensus Score:**

Computes agreement percentage among individual top predicted classes:

- **Unanimous Consensus (100% Agreement):** High clinical confidence across all backbones.
- **Majority Consensus (66.7% Agreement):** Majority model concurrence.
- **Inter-Model Disagreement (< 66.7% Agreement):** Automatically **flagged for radiologist expert review**.

2.3 Epistemic Uncertainty Estimation Engine (`modules/uncertainty_engine.py`)

Medical AI predictions require statistical confidence quantification. TumorAI XAI implements **Monte Carlo (MC) Dropout stochastic inference**:

- **Mechanism:** Retains active Dropout layers ($p = 0.2$ to 0.5) during inference across $N = 10$ stochastic forward passes for a single MRI image.
- **Uncertainty Metrics:**
- **Prediction Variance:** Variance of probabilities across the N passes. High variance indicates model ambiguity.

- **Predictive Entropy:** Shannon entropy computed over mean prediction probabilities:

$$H(Y|X) = -\sum_c \bar{P}_c \log_2(\bar{P}_c)$$

- **Mean Confidence:** Mean top-class probability over all Monte Carlo passes.

3. Explainable AI (XAI) & Spatial Validation

To transform black-box neural networks into transparent clinical tools, TumorAI XAI provides three visual explanation techniques alongside quantitative spatial validation metrics.

3.1 Visual Explanation Techniques (`modules/xai_engine.py`)

1. Grad-CAM (Gradient-weighted Class Activation Mapping)

- Computes partial derivatives of the target class score y^c with respect to feature maps A^k in the target layer (`block5_conv3`).
- Generates a coarse localization map highlighting important regions used by the model for classification, overlaid as a OpenCV JET colormap on the MRI.

2. Grad-CAM++

- Incorporates second and third-order gradients to provide superior localization for multiple instance lesions and finer spatial detail.

3. LIME (Local Interpretable Model-agnostic Explanations)

- Segments the MRI image into superpixel components using Quickshift / SLIC algorithms.
- Perturbs superpixels and fits a local interpretable ridge linear model to evaluate positive and negative superpixel contributions to the diagnostic output.

4. Computer Vision ROI Extraction

- Employs OpenCV contour thresholding (`localization_method="opencv_contour"`) to generate a clean binary mask of the suspected tumor Region of Interest (ROI).
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3.2 Spatial Validation & Attribution Metrics (`modules/xai_metrics.py`)

Evaluates the spatial alignment between generated XAI saliency heatmaps (thresholded at top intensity percentiles) and the extracted ROI ground-truth contour:

- **Intersection over Union (IoU):**

$$\text{IoU} = \frac{|M_{\text{XAI}} \cap M_{\text{ROI}}|}{|M_{\text{XAI}} \cup M_{\text{ROI}}|}$$

- **Dice Similarity Coefficient (DSC):**

$$\text{DSC} = \frac{2 \cdot |M_{\text{XAI}} \cap M_{\text{ROI}}|}{|M_{\text{XAI}}| + |M_{\text{ROI}}|}$$

4. Core Application Features & Services

4.1 Patient & Study Management (`services/patient_service.py`, `services/study_service.py`)

- Full patient lifecycle management including demographics, Medical Record Number (MRN), Date of Birth, Gender, Blood Type, Contact Details, and Clinical History.
- Clinical Imaging Study creation grouping multiple MRI scans, slice series, and longitudinal diagnostic tracking over time.

4.2 Medical Imaging & DICOM Processor (`services/dicom\_service.py`)

- Native `.dcm` (DICOM) file parsing using `pydicom`.
- Metadata extraction: Patient Name, Patient ID, Modality (MR), Study Date, Manufacturer, Window Center/Width, Pixel Spacing, Slice Thickness.
- Automated extraction and windowing conversion of raw pixel arrays into high-contrast PNG/JPEG images for browser rendering and neural inference.

4.3 LLM Clinical Summary Generator (`services/gemini\_service.py`)

- Integrates Google Gemini AI (`gemini-1.5-flash` / `gemini-pro`) to analyze inference results, XAI heatmaps, ensemble consensus, and uncertainty metrics.
- Automatically generates structured radiologist diagnostic summaries containing:
 - Clinical Impression
 - Key Findings & Anatomic Location
 - Uncertainty & Disagreement Analysis
 - Recommended Next Steps / Follow-up MRI Protocol

4.4 HL7 FHIR R4 Interoperability (`services/fhir\_service.py`)

- Transforms local scan predictions and patient study records into standardized HL7 FHIR R4 JSON payloads:
- **FHIR `DiagnosticReport`**: Structured clinical diagnostic report resource.
- **FHIR `Observation`**: Quantitative observation resource recording diagnostic probability, confidence score, and entropy.
- Enables seamless integration with Hospital Information Systems (HIS) and Electronic Health Record (EHR) platforms (e.g., Epic, Cerner).

4.5 PDF Diagnostic Report Builder (`services/report\_service.py`)

- Generates publication-quality multi-page PDF medical reports using ReportLab.
- Embedded elements: Hospital header, patient demographics, raw MRI scan, Grad-CAM heatmap overlay, ROI boundary, ensemble probabilities, uncertainty bounds, Gemini AI clinical notes, and radiologist digital signature block.

4.6 Automated Email Dispatcher (`services/email\_service.py`)

- Secure SMTP email service for dispatching generated PDF diagnostic reports directly to attending physicians or patients as email attachments.

4.7 Enterprise Security & Audit Trail (`services/audit\_service.py`)

- Comprehensive audit logging capturing every platform action: scan uploads, inference executions, report generations, patient updates, and record deletions.
 - Records user ID, IP address, action type, resource ID, timestamp, and metadata for HIPAA/GDPR compliance audits.
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5. Database Schema & Storage Architecture

5.1 Supabase PostgreSQL Database Schema (`supabase/migrations/`)

Table Name	Description	Key Columns
profiles	User accounts and radiologist profiles	id, email, full_name, role, created_at
patients	Patient demographic records	id, mrn, first_name, last_name, dob, gender, blood_type, medical_history
studies	Imaging studies linking scans to patients	id, patient_id, study_uid, study_date, description, modality, status
scans	MRI & DICOM scans with inference results	id, study_id, image_path, roi_path, gradcam_path, gradcam_pp_path, lime_path, prediction, confidence, uncertainty_variance, entropy, consensus_status
reports	Diagnostic reports and summaries	id, scan_id, summary, ai_findings, pdf_report_path, status, created_at
study_notes	Radiologist notes attached to studies	id, study_id, author_id, note_text, created_at
audit_logs	Security and operational activity logs	id, user_id, action, resource_type, resource_id, ip_address, details, timestamp

5.2 Supabase Private Storage Buckets (`services/storage\_service.py`)

- **mri-scans Bucket** (Private): Holds raw MRI images, extracted DICOM frames, ROI masks, and XAI heatmaps (Grad-CAM, Grad-CAM++, LIME).
- **patient-reports Bucket** (Private): Holds generated PDF diagnostic reports.
- **Short-Lived Signed URLs**: Frontend requests signed URLs on-demand with a default expiration of **300 seconds (5 minutes)**, preventing unauthorized direct asset access.

6. Web Interface & User Experience (`templates/index.html`)

The frontend is a modern single-page dashboard designed with Vanilla JavaScript, custom CSS tokens, glassmorphism, dynamic transitions, and responsive tab navigation:

- Executive Dashboard:** System metrics, total patient counts, scan volume, tumor distribution charts, and quick system status indicator.
- Patient Management Hub:** Searchable patient table, MRN lookup, new patient registration modal, and patient medical profile views.
- MRI & DICOM Upload & Real-Time Inference:** Drag-and-drop file uploader (supporting .png, .jpg, .jpeg, .dcm), model architecture dropdown (VGG16, ResNet50, EfficientNetB0, or Ensemble), pre-inference quality audit status, and real-time progress visualization.
- XAI Visual Analytics Suite:** Multi-pane synchronized visual comparison viewer:
 - Original Scan
 - Grad-CAM Heatmap
 - Grad-CAM++ Localization Map
 - LIME Superpixel Patch Attribution

- OpenCV ROI Mask
 - Counterfactual Sensitivity Map
 - IoU & Dice Score spatial metrics sidebar
5. **Interactive DICOM & 3D Tumor Viewer:** Canvas rendering with brightness/contrast adjustments, zoom/pan controls, DICOM tag header inspector, orthogonal slice planes (Axial, Coronal, Sagittal), and 3D surface mesh rendering.
6. **AI Reliability & Model Consensus Matrix:** Displays overall AI reliability score (0-100%) and backbone agreement matrix (VGG16 vs ResNet50 vs EfficientNetB0).
7. **Tumor Segmentation & Volumetric Analytics:** Displays 3D volume in cm^3 , max linear diameter in cm , and boundary overlays.
8. **Longitudinal Growth Tracker:** Interactive timeline chart depicting volume delta (ΔV), monthly growth velocity, and RANO/RECIST treatment response categories.
9. **Study-Grounded Clinical AI Copilot:** Interactive Q&A; chat assistant grounded strictly in active patient study evidence.
10. **Reports & Gemini AI Clinical Synthesis:** One-click Gemini AI diagnostic summary generation, rich-text radiologist note editor, PDF report preview/download, SMTP email sender, and FHIR export.
11. **Compliance Audit Log Viewer:** Searchable, filterable audit log timeline.

7. Directory Structure Overview

```
brain tumor/
├── main.py                # Primary Flask Application & REST API Endpoints
├── requirements.txt       # Dependencies (TensorFlow, Supabase, PyDICOM, etc.)
├── Dockerfile            # Docker Container Configuration
├── Procfile              # Heroku / Cloud Deployment Process File
├── README.md             # Quickstart & Manual Supabase Setup Guide
├── PROJECT_DOCUMENTATION.md # Comprehensive Architecture & Features Manual (This File)
├── .env.example          # Environment Variable Template
├──
├── models/               # Pre-trained Model Weights
│   ├── vgg16_best.h5     # VGG16 Weights
│   ├── resnet50_best.h5  # ResNet50 Weights
│   ├── efficientnetb0_best.h5 # EfficientNetB0 Weights
│   └── model.h5          # Primary Classifier Backup Weights
├──
├── modules/              # Core ML, XAI, Segmentation & Uncertainty Engines
│   ├── ensemble_engine.py # Multi-Model Voting & Consensus Calculation
│   ├── model_trainer.py   # Model Building, Transfer Learning & Fine-tuning
│   ├── uncertainty_engine.py # Monte Carlo Dropout Uncertainty & Entropy
│   ├── xai_engine.py      # Grad-CAM, Grad-CAM++, LIME & Contour ROI
│   ├── xai_metrics.py     # IoU & Dice Similarity Coefficient Calculations
│   ├── segmentation_engine.py # Automatic 3D Tumor Segmentation & Volume (cm³)
│   ├── quality_checker.py  # Pre-Inference AI Data Quality & OOD Audit
│   └── counterfactual_engine.py # Counterfactual XAI & Sensitivity Analysis
├──
├── services/             # Enterprise Backend Business Services
│   ├── supabase_client.py # Supabase SDK Client Initialization
│   ├── storage_service.py  # Private Bucket Uploads & Signed URL Generators
│   ├── patient_service.py  # Patient CRUD Operations
│   ├── study_service.py    # Clinical Study Operations
│   ├── scan_service.py     # MRI Scan Persistence & Retrieval
│   ├── report_service.py   # ReportLab PDF Generation
│   ├── dicom_service.py    # DICOM Parsing & Conversion
│   ├── gemini_service.py   # Google Gemini AI LLM Diagnostics Integration
│   ├── fhir_service.py     # HL7 FHIR R4 Serialization (DiagnosticReport, Observation)
│   ├── email_service.py    # SMTP Email PDF Dispatching
│   ├── audit_service.py    # Compliance Audit Logging
│   ├── analytics_service.py # Dashboard Analytics & Statistics Computation
│   └── longitudinal_service.py # Longitudinal Tumor Growth & RANO Response Monitor
```

```

■   ■■■ copilot_service.py           # Study-Grounded Clinical AI Copilot Q&A
■   ■■■ data_migration_service.py    # Database Migration & Sync Utilities
■
■■■ supabase/                        # Database Schemas & Migrations
■   ■■■ migrations/
■       ■■■ 20260829000000_init_supabase_schema.sql # Database DDL & Schema Setup
■
■■■ templates/                      # Frontend Views
■   ■■■ index.html                  # Full Enterprise SPA Frontend Interface
■   ■■■ index1.html                 # Legacy Frontend Backup
■
■■■ static/                          # Static Assets (CSS, JS, Images)
■■■ tests/                          # Automated Unit & Integration Tests
■   ■■■ test_supabase_xai_system.py # Comprehensive End-to-End System Tests
■   ■■■ test_new_features.py        # Advanced Intelligence Suite Automated Unit Tests
■■■ uploads/                        # Temporary Local File Buffer

```

8. How to Setup and Run

8.1 Environment Setup

1. Create a Python Virtual Environment:

```
python -m venv venv
source venv/bin/activate # On Windows: .\venv\Scripts\activate
```

2. Install Required Dependencies:

```
pip install -r requirements.txt
```

8.2 Configuration (`.env`)

Create a `.env` file in the project root:

```

FLASK_ENV=development
PORT=5000

# Supabase Credentials
SUPABASE_URL=https://your-project-ref.supabase.co
SUPABASE_ANON_KEY=your-anon-public-key
SUPABASE_SERVICE_ROLE_KEY=your-service-role-key

# Supabase Storage Buckets
SUPABASE_MRI_BUCKET=mri-scans
SUPABASE_REPORT_BUCKET=patient-reports
SIGNED_URL_TTL_SECONDS=300

# Google Gemini API Key
GEMINI_API_KEY=your-gemini-api-key

# SMTP Configuration (Optional for PDF Emailing)
SMTP_SERVER=smtp.gmail.com
SMTP_PORT=587
SMTP_USERNAME=your-email@gmail.com
SMTP_PASSWORD=your-app-password

```

8.3 Launching the Application

```
python main.py
```

Navigate to <http://127.0.0.1:5000> in your browser.

8.4 Running System Tests

```
python -m unittest discover -s tests -p "test_*.py"
```

9. Summary of Guarantees

- **Zero Breaking Edits:** All pre-existing features, UI elements, ML classification routines, and database tables remain 100% functional and preserved.

- **Single Master Reference File:** All technical architectural specifications, feature workflows, and setup guidelines are encapsulated within this file

[\[PROJECT_DOCUMENTATION.md\]](#)(file:///c:/Users/Purba%20Hanra/brain%20tumor/PROJECT_DOCUMENTATION.md).